

Pokhara University
Faculty of Science and Technology

Course No.: xxx xxx

Course title: Basic Electrical Engineering (3-1-2)

Nature of the course: Theory & Practical

Level: Undergraduate

Full marks: 100

Pass marks: 45

Time per period: 1 hour

Total Periods: 45

Program: BE

1. Course Description

This course covers the various concepts of electrical circuits, theorems as well as the concepts of electrical machines. This course emphasizes on fundamental concept, principles and properties of electrical circuits, circuit parameters and its application. It also covers the concepts of DC and AC electrical circuit analysis and electrical machine.

2. General Objective

The general objectives of this course are:

- To acquaint the students with electric circuits (A.C. & D. C), steady state behavior of single phase and three phase A.C. electrical circuits.
- To make the students able to operate, distinguish and use electrical devices and machines.

3. Methods of Instruction

Lecture, Discussion, Readings, Lab works, Project works

4. Contents in Detail

Specific Objectives	Contents
Describe about the generation, transmission, distribution and use of electrical energy	Unit I: Introduction (2 hrs) - Role of electricity in modern society, - Energy sources and production, - Generation, transmission and distribution of electrical energy, - Consumption of electricity
Apply basic rules and theorem in circuit analysis	Unit II: DC Circuit Analysis (15 hrs) - Circuits concepts (lumped and distributed parameters), linear and nonlinear parameter, passive and active circuits, - Circuit elements (Resistance, capacitance and inductance), their properties and characteristics in a geometrical and hardware aspects, - Color coding, Series of parallel compilation of resistances, Equivalent resistance and its calculation, star-delta transformation, - Concept of power, energy and its calculations, - Short and open circuit, - Ideal and non-ideal sources, source conversion,

	- Voltage divider and current divider formula, - Kirchhoff's current and voltage laws, - Nodal method and mesh method of network analysis (without dependent source), - Network theorem (Superposition, Thevenin's, Norton's, maximum power transfer theorem)
- Analyze generation of single-phase alternating quantities and its characteristics - Analyze the steady state behavior of single-phase ac electrical circuits	Unit III: Single Phase AC Circuits Analysis (10 hrs) - Generation of EMF by electromagnetic induction, Generation of alternating voltage, - Sinusoidal functions-terminology (phase, phase angle, amplitude, frequency, peak to peak value), average values and RMS or effective value of any types of alternating voltage or current waveform, - Phase algebra, power triangle, impedance triangle, steady state response of circuits (RL, RC, RLC series and parallel) and concept about admittance, impedance, reactance and its triangle), instantaneous power, average real power, reactive power, power factor and significance of power factor, - Resonance in series and parallel RLC circuit, bandwidth, effect of Q factor in resource
- Analyze generation of poly phase alternating quantities and its characteristics - Analyze the steady state behavior of three phase ac electrical circuits. - Describe the measurement of three phase power	Unit IV: Poly-phase AC Circuit Analysis (6 hrs) - Generation of three phase alternating quantity and Concept of a balanced three phase supply, - Differences between single phase and three phase system, - Star and delta connected supply and load circuits., Line and phase voltage/current relations, power measurement, - Concept of three phase power and its measurement by single and two wattmeter methods
- Analyze performance and operation of single phase two winding transformer - Analyze performance and operation of DC machine - Analyze performance and operation of some AC machines	Unit V: Electrical Machines (12 hrs) - Differences and similarities between electric circuit and magnetic circuit - Transformers: Principle of operations, features, equivalent circuits, efficiency & regulation, open circuit & short circuit tests of single phase two winding transformer. - DC generator: Construction features, working principles, basic characteristics - DC motors: Performance & operation, basic characteristics, speed control & selection of motors - AC machines: Single phase and three phase induction motors (working principles, construction features and uses)

5. Laboratory Work

- a. To measure current, voltage and power across the passive components.
- b. To verify Kirchhoff's Current Law (KCL) & Kirchhoff's Voltage Law (KVL)
- c. To verify Thevenin's Theorem.
- d. To verify maximum power transfer theorem.
- e. To verify superposition theorem.
- f. To measure three phase power by using two wattmeter
- g. To determine efficiency and voltage regulation of a single-phase transformer by direct loading.
- h. To study open circuits & short circuits tests on a single-phase transformer
- i. To study the speed control of dc shunt motor by.
 - i. Varying the field current with armature voltage held constant field control.
 - ii. Varying the armature voltage with field current held constant armature control.
- j. To study open circuits and load test on a dc shunt generator (separately excited)
 - i. To determine magnetization characteristics
 - ii. To determine V-I characteristics of a dc shunt generator

6. List of Tutorials

The various tutorial activities that suits your course should cover all the content of the course to give students a space to engage more actively with the course content in the presence of Instructor/professor. The following tutorial activities of 15 hrs. should be conducted to cover the content of this course:

1. Discussion-based Tutorials: (1 hr)
2. Numerical discussion on DC circuit analysis (star/delta conversion, mesh analysis, nodal analysis, superposition theorem, Thevenin's theorem, Norton's theorem and Maximum power transfer theorem. (5 hrs)
3. Numerical discussion on single phase ac circuit (3 hrs)
4. Analysis and numerical solution of three phase ac circuits. (2 hrs)
5. Numerical problems discussion on single phase Transformer, dc motor, dc generator. (4 hrs)

7. Evaluation system and Students' Responsibilities:

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, term-exams, lab reports and projects etc. The tabular presentation of the internal evaluation is as follows:

External Evaluation	Marks	Internal Evaluation	Weight	Marks
Semester-End examination	50	Theory		30
		Attendance & Class Participation	10%	
		Assignments	20%	
		Presentations/Quizzes	10%	
		Term exam	60%	
		Practical		20
		Attendance & Class Participation	10%	
		Project Report	10%	
		Viva	20%	

		Exam	60%	
		Total Internal		50
Full Marks: 50 + 50 = 100				

Student Responsibilities

Each student must secure at least 45% marks in internal evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Books and References

Text Books

1. Boylested, Albert "Introduction of Electric circuit" Prentice Hall of India Private Limited, New Delhi
2. Tiwari, S.N, "A first course of electrical engineering" att. Wheeler & Co. Ltd. Allabhad.

References

1. Thereja B. L & Thereja A. K. "A text book of Electrical Technology, S Chand Publication.
2. Jain& Jain "ABC of Electrical Engineering"

Pokhara University
Faculty of Science and Technology

Course No.: xxx xxx
Course title: Calculus I (3-2-0)
Nature of the course: Theory
Level: Undergraduate

Full marks: 100
Pass marks: 45
Time per period: 1 hour
Total Periods: 45
Program: BE

1. Course Description

This course is designed for developing competency of the students on the fundamental concepts, principals and applications of differential and integral calculus for solving engineering problems. It is equipped with differential calculus, integral calculus and ordinary differential equations. The review part of the content is based on previous learning in the school level. The course will be delivered through lecture method, assignment on practically base engineering problems and class tests.

2. General Objectives

The course is designed with the following general objectives:

- To acquaint the students with applications of differential and integral calculus in engineering.
- To expose the students with the application of differential equations for modeling engineering problems.

3. Methods of Instruction

Lecture, tutorials and assignments

4. Contents in Detail

Specific objectives	Contents
• Explain the importance of limit and continuity in differential problems and use Leibnitz theorem to evaluate higher derivatives.	Unit I: Limit Continuity and Derivatives (5 hrs) 1.1 Introduction 1.2 Limit, continuity and differentiability 1.3 Higher order derivatives by Leibnitz method.
• Apply derivatives in mean value theorem, series expansion, asymptotes and trace curve for the given function.	Unit II: Applications of Derivatives (8 hrs) 2.1 Mean value theorems: Rolle's theorem, Lagrange's Theorem (Geometrical interpretation and verification) and applications 2.2 Higher order mean value theorem: Taylor's Series, Maclaurin's Series expansion of function. 2.3 Asymptotes to Cartesian curves up to four degrees. 2.4 Curve tracing in Cartesian form and parametric form 2.5 Curvature
• Evaluate Proper and improper integrals.	Unit III: Integral Calculus (6 hrs) 3.1 Introduction 3.2 Review on Indefinite Integral and fundamental theorem

	of integral calculus. 3.3 Definite integral and its properties 3.4 Improper Integrals; comparison test. 3.5 Reduction formula, Beta Gamma functions
Evaluate arc length, area, and volume through integration.	Unit IV: Application of Integral (6 hrs) 4.1 Application of integrals for finding area beneath a curve and between two curves and arc length 4.2 Surface and volume of solid of revolution in the plane for Cartesian and parametric curves.
Understand and compute partial derivatives with the concept of total differentials.	Unit V: Partial Differentiation (3 hrs) 5.1 Introduction 5.2 Partial Derivatives 5.3 Homogeneous function and Euler's theorem for the function of two and three variables 5.4 Total Derivatives and Differentiation of Implicit functions.
Define extreme value and compute its value for two and three variables through partial derivatives.	Unit VI: Application of Partial Differentiation (4 hrs) 6.1 Extrema of functions of two and three variables. 6.2 Lagrange's method of undetermined Multipliers (up to 2 multipliers)
Able to solve first order differential equations.	Unit VII: First Order Ordinary Differential Equations (6 hrs) 7.1 Review of separable, homogeneous and exact differential equation with engineering applications 7.2 Linear, Bernoulli equation and Riccati's equation with engineering application. 7.3 Mathematical modeling of engineering problems using first order equation.
Solve the second order differential equations in relation to engineering problems.	Unit VIII: Second Order Ordinary Differential Equations (7 hrs) 8.1 Second order Homogeneous ODE with constant and variable coefficients, Euler-Cauchy equation. 8.2 Existence and uniqueness of solutions, Wronskian and general solutions for solving ODE. 8.3 Non-homogeneous second order ODE and Solution by undetermined coefficients and variation of parameters and engineering application

Note: The figures in the parentheses indicate the approximate periods for the respective units.

5. List of Tutorials

Tutorial work covers the work to be done in tutorial. This will enable the students to compute the mathematics problem under the supervision of the course leader. The major tutorial works are as follows:

- Problems on Limit and continuity (1 hr)
- Show that differentiability implies continuity but the converse may not be true (1 hr)
- Evaluation of higher order derivatives by Leibnitz method (1 hr)

4. Problems in Mean value theorems: Rolle's theorem, Lagrange's Theorem (1 hr)
5. Expand the functions through Taylor's Series, and Maclaurin's Series (2 hrs)
6. Evaluation of Asymptotes to Cartesian curves. (2 hrs)
7. Trace Curve for the equations in Cartesian form and parametric form (2 hrs)
8. Problems in Curvature (1 hr)
9. Evaluation of Indefinite Integrals, Definite integrals, Improper Integrals; (2 hrs)
10. Deduce Reduction formula, and solve problems related to Beta Gamma functions (2 hrs)
11. Evaluation of area, arc length, Surface and volume of solid of revolution in the plane for Cartesian and parametric curves. (3 hrs)
12. Prove Euler's theorem for the function of two and three variables (1 hr)
13. Calculate total derivatives and differentiation of Implicit functions. (1 hr)
14. Evaluation of Extrema of functions of two and three variables and Lagrange's method of undetermined Multipliers (up to 2 multipliers) (2 hrs)
15. Solution of separable, homogeneous and exact differential equation Linear, Bernoulli equation and Riccati's equation with engineering applications (2 hrs)
16. Mathematical modeling of engineering problems using first order equation. (1 hr)
17. Solve second order homogeneous ODE with constant and variable coefficients, Euler-Cauchy equation. (3 hrs)
18. Solve non-homogeneous second order ODE by undetermined coefficients and variation of parameters in engineering application (2 hrs)

6. Evaluation System and Students' Responsibilities

Evaluation System

Internal evaluation is done as follows:

External Evaluation	Marks	Internal Evaluation	Weight	Marks
Semester-End Examination	50	Attendance & Class Participation	10%	
		Assignments	20%	
		Presentations/Quizzes	10%	
		Term exam	60%	
		Total Internal		50
Full Marks: 50 + 50 = 100				

Students' Responsibilities

Each student must secure at least 45% marks in internal evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

7. Prescribed Books and References

Text Books

1. Stewart, J. *Calculus, Early Transcendental*. India; Cengage Learning.
2. Kreyszig, E. *Advance Engineering Mathematics*, New Delhi: John Wiley and Sons Inc.

References

1. Thomas, G. & Finney, R. *Calculus and Analytical Geometry*. New Delhi: Narosa Publishing House.
2. Mishra, P., Mishra, R., Mishra, V. P., & Mishra, M. *Advance Engineering Mathematics*. New Delhi: V. P. Mishra Publication.
3. Dass, H. K. & Verma R. *Higher Engineering Mathematics*. New Delhi: S Chand Publishing.

Pokhara University
Faculty of Science and Technology

Course No.:
Course Title: Communication Techniques (2-2-0)
Nature of the Course: Theory
Year: First
Level: Undergraduate
Credits: 2

Full Marks: 100
Pass Marks: 45
Time per Period: 1 hour
Total Periods: 30

Program: BE

1. Course Description

This course is designed to offer a comprehensive introduction to first-year engineering students to strategies that will help them create effective technical documents and presentations.

It covers the entire gamut of technical communication in thirteen major parts, namely, thinking about the audience, purpose, and genre; ethical issues; researching; technical writing; designing documents; visual communication; communication in the workplace; writing proposals; reporting information; writing how-to documents; reporting document usability; taking communication online; and presenting communication orally.

This course takes a rhetorical approach to technical communication. This means that instead of setting up a list of rules that students should apply uniformly to all writing situations, this course introduces them to the bigger picture of how the words they write can affect the people intended to use them. By understanding who the readers or users are and what they need in a technical document, students can adapt their knowledge to their situations to provide them with what they need and this will facilitate how to educate themselves as well as facilitate the learning of others. There are also several different kinds of exercises and assignments. in-class exercises are short assignments intended to be done by students in class in 15 or 20 minutes. They ask students to use the main ideas discussed in the previous section and to think critically about those ideas. There are also assignments and major projects, and these group activities further learning through opportunities to work with others, to discuss the course content, and to hear others' bright ideas that might spark greater creativity overall.

2. General Objectives

The objectives of the course are to enable students to

- practice technical writing strategies
- apply the concepts that they need to make good decisions about how to write a document
- guide them in developing a good working draft
- test their document with members of the target user group and receive constructive feedback
- present usable information on a specialized (and usually technical) subject
- write technical proposals, reports, and documents
- make oral technical presentations
- communicate in the workplace.

3. Methods of Instruction

In this course the idea is that students should read the chapter prior to class and, when they get to class, the teacher can quickly summarize for them the main points to which they should pay attention. Then the teacher can assign an exercise that will reinforce the theoretical concept and get students started working with it. There are also several different kinds of exercises and assignments: in-class exercises are short assignments intended to be done by students in class in 15 or 20 minutes, and here students use the main ideas discussed in the previous section and think critically about those ideas; lab assignments are slightly longer than in-class exercises, but they are still intended to be completed (or at least worked on) during a class meeting; the third type of assignment is the major project, which is a longer, more formal assignment that has students consider the main concepts from the chapter (or chapters) to produce an effective example of one of the main genres of technical communication. The teacher can assign the major project at the beginning of a particular unit of study and have it due the following week or at the end of the term, depending on the teacher's course plan and schedule. It is also important to have the students work collaboratively on the in-class and lab assignments. Depending upon the teacher's course goals, they may decide to cover the chapters in the order in which they are presented in the syllabus but they can use the chapters in the order that suits their students' needs and the assignment schedule.

4. Contents in Detail

The course is divided into eight units of two weeks each.

Weeks	Specific Objectives	Course Contents
1-2	Introducing the basic concepts of audience, purpose, and genre; Introducing ethical issues and conducting research	Thinking about audience, purpose, and genre; Leading and misleading the reader: - ethics at work - ethics for students - how is ethics related to technical communication? - researching technical subjects
3-4	Writing for the workplace; Style in technical prose	Writing email and letters for the workplace; - writing messages: email, memos, letters - writing messages: informative, positive, negative, persuasive - how is ethics related to technical communication Writing technical prose: - clarity, cohesion, conciseness, parallelism
5-6	Designing technical documents; Incorporating visuals into technical documents	Designing documents and page layout; Communicating through visuals
7-8	Writing effective proposals; Writing instructional documents	Writing winning proposals; Writing how-to documents: - instructions, procedures, and manuals
9-10	Testing instructional documents;	Testing and reporting document usability;

	Reporting technical information, part 1	Reporting technical information: - recommendation reports
11-12	Reporting technical information, part 2; reporting technical information, part 3 and accessing technical documents online	Reporting technical information: - white papers or information reports; Taking technical communication online: - sharing documents electronically
13-14	Writing documents to be used online; Status or progress reports	Taking technical communication online: - writing online documents; Reporting technical information: - status or progress reports
15-16	Presenting technical information orally; Students present some aspect of their major projects	Oral presentation of technical reports; Student presentations of major project

5. Evaluation System and Students' Responsibilities

Evaluation System

In addition to the formal exam(s), the internal evaluation of a student may consist of quizzes, assignments, project work, class participation, etc. The tabular presentation of the internal evaluation is as follows.

External Evaluation	Marks	Internal Evaluation	Weight	Marks
Semester-End Examination	50	Attendance & Class Participation	10%	
		Assignments	20%	
		Presentations/Quizzes	10%	
		Term exam	60%	
		Total Internal		50
Full Marks: 50 + 50 = 100				

6. Student Responsibilities

Each student must secure at least 45% marks in internal evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

7. Prescribed Books and References

Text Book

Graves, H., & Roger, G. *A Strategic Guide to Technical Communication*. 2nd ed., London: Eurospan Group, 2012.

References

- Greenlaw, R. *Technical Writing, Presentation Skills, and Online Communication: Professional Tools and Insights*. IGI Global, 2012.
- Gurak, L. J. & John, M. L. *Strategies for Technical Communication in the Workplace*. 2nd ed., Pearson, 2013.
- Kmiec, D. & Bernadette, L. *The IEEE Guide to Writing in the Engineering and Technical Fields*. Wiley, 2017.
- Markel, M. *Technical Communication*. 11th ed., Bedford/St. Martins, 2015.
- Mirel, B. & Rachel, S. eds. *Reshaping Technical Communication: New Directions and Challenges for the 21st Century*. Lawrence Erlbaum, 2002.

Pokhara University
Faculty of Science and Technology

Course No.: xxx xxx
Course title: Computer Workshop
Nature of the course: Practical
Year: First
Level: Bachelor

Full marks: 100
Pass marks: 45
Time per period: 1 hour
Total periods: 30
Program: BE

1. Course Description

This course provides the knowledge and hands-on skills of computer hardware, software, computer networking and enables students to identify and rectify the onboard computer hardware, software and network related problems. Students will be able to understand the hardware specifications for the operating system and various application programs.

General Objectives

The general objectives of this course are:

- To familiarize the students with the computer systems hardware, basic practical works and computer networking concepts
- To make the students competent in install/update operating system and various application software, manage data backup and restore operations on a computer system

2. Methods of Instruction

Specific instructional techniques: Lab works, Project works

3. Contents in Detail

Specific Objectives	Contents
- Familiar with computer hardware and accessories	Unit 1: Introduction to Desktop/Workstation Assembly (2) 1.1 Demonstration of computer hardware and peripheral devices: Processors, Motherboard, Memory, Storage devices, I/O Devices. 1.2 Demonstration of Power Supply-SMPS, Internal cablings and Ports.
- Assemble and Disassemble PC with power supply, internal cabling, Motherboard and PCI Device	Unit 2: Assembling Desktop Computer (2) 2.1 Assembling steps and precautions. 2.2 Setting of motherboards, memory, hard disk, processors, CDROM, SMPS and other devices.
- Assemble Laptop power supply, internal cabling, Motherboard and PCI Devices.	Unit 3: Assembling Laptop (2) 3.1 Assembling steps and precautions. 3.2 Demonstration of motherboards, memory, hard disk, processors, CDROM, SMPS and other devices.

- Install OS (Linux and Windows) in Desktop/Laptop	Unit 4: Laptop/Desktop OS Installation (6) 4.1 BIOS setting, firmware types, BOOT configuration. 4.2 Hard disc partitioning. 4.3 OS installation (Linux, windows). 4.4 User account management 4.5 Printer, Scanner installation.
- Visualize computer network and familiar with cabling and LAN setup - Recognize Wireless LAN and AP configuration	Unit 5: Networking and Internet Setup (5) 5.1 Introduction to Computer Network, Network topologies, wired and wireless networking media. 5.2 Network cabling, cable types and connectors. 5.3 Basic wireless AP configuration.
- To understand different types of Application/Utility software. - To be able to Install and Update of Antivirus	Unit 6: Software Installation (5) 6.1 Installation of application and utility software. 6.2 Update of firmware patches, 6.3 Virus scanning.
- To know about basic concept server and Setup Server of Mail/Data/Domain/FTP - To Manage Data backup and Restore operation from server and Update in Application Software - Uses of FTP/SCP Client tool to upload/download files to FTP server.	Unit 7: Server Installation and Backup (3) 7.1 Introduction to server. 7.2 IP Address setting. 7.3 Study of different server (Email, Data, Domain, FTP).
- Able to know about Multimeter, Soldering iron, Hot gun. - Able to understand the troubleshooting tools and utilities.	Unit 8: Hardware and Software Troubleshooting (5) 8.1 Basic Repair and Maintenance of Desktop/Laptop. 8.2 Replacement of Passive Components. 8.3 Uses of basic software troubleshooting commands. 8.4 Uses of trouble shooting tools and utilities.

4. Evaluation system and Students' Responsibilities

Evaluation System

The evaluation of a student may consist of assignments, attendance, test-exams, term-exams, lab reports and projects etc. The tabular presentation of the evaluation is as follows:

External Evaluation	Marks	Internal Evaluation	Weight	Marks
Semester-End examination	NA	Attendance & Class Participation	10%	
		Assignments	10%	
		Quizzes/MCQ	10%	

		Class Test	20%	50
		Assessment Exam	50%	
		Internal Final Exam	100%	50
Full Marks: 50 + 50 = 100				

Student Responsibilities

Each student must secure at least 45% marks in evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear the Final Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

5. Prescribed Books and References

1. Winn L. Rosch, Hardware Bible, 6th Edition.
2. Sanjay K Bose, Hardware and Software of Personal Computers.
3. V. Rajaraman and Neeharika Adabala, Fundamental of Computers.

Pokhara University
Faculty of Science and Technology

Course No.: xxx xxx
Course title: Digital Logic (3-1-2)
Nature of the course: Theory & Practical
Level: Undergraduate

Full marks: 100
Pass marks: 45
Time per period: 1 hour
Total Periods: 45
Program: BE

1. Course Description

This course covers the various concepts of digital logic systems. This course emphasizes on fundamental concept, principles and properties of Boolean algebra and its application in simplification, circuit analysis and gate implementation. It covers the use of flip flops in the design of synchronous and asynchronous sequential logic circuits. It also covers the ALU design.

2. General Objectives

This course provides basic knowledge of logic systems, introduces basic tools to design various digital logic circuits and enables the students to design a basic digital computer.

3. Methods of Instruction

Lecture, discussion, readings, Lab works, Project works

4. Contents in Detail

Specific Objectives	Contents
- Familiarize and compare analog and digital signal and system. - Familiarize and use the information representation in any number system.	Unit 1: Introduction (2 hrs) - Analog and digital signal - Analog and digital system - Numerical representation - Digital number system
- Familiarize with binary number systems and their conversions. - Understand the different codes to represent information.	Unit 2: Number Systems and Codes (6 hrs) - Number systems - Decimal - Binary - Octal - Hexadecimal - Number system conversion - Complements (radix and diminished-radix) - Subtraction using complements - Binary coding systems - Weighted codes (BCD, 8 4 -2 -1, and 2 4 2 1) - Non-weighted codes (Excess-3 and Gray) - Alphanumeric and instruction codes

- To trace the simplification process using Boolean Algebra. - To implement the simplified functions using logic gates.	Unit 3: Boolean Algebra and Logic Gates (4 hrs) 3.1 Boolean algebra (definition, properties, postulates and theorems) 3.2 Logic gates, truth tables and Boolean function 3.3 Duality principle and complements 3.4 Gate implementation 3.5 Universality of NAND and NOR gates
- Simplify the Boolean function using map method. - Help to identify and use don't care conditions while simplifying functions.	Unit 4: Simplification of Boolean Function (5 hrs) 4.1 Venn diagram 4.2 Canonical forms and standard forms 4.3 Karnaugh map up to 5 variables 4.4 Minimum realization 4.5 Don't care conditions 4.6 Simplification in SOP and POS using K-map
- Describe how to design various combinational logic circuits. - Help to analyze the designed combinational circuits.	Unit 5: Combinational Circuit (4 hrs) 5.1 Design procedure 5.2 Adder and subtractor 5.3 Code conversion 5.4 Analysis procedure 5.5 NAND and NOR implementation 5.6 Multilevel NAND and NOR gates 5.7 Parity generator and checker
- Describes different generations of IC technology. - Describe different MSI and LSI components, their design, internal logic diagram, operation and implementation.	Unit 6: MSI and LSI Design (6 hrs) 6.1 Introduction to Integration technology 6.2 Parallel adder and subtractor 6.3 Decimal / BCD adder 6.4 Magnitude comparator 6.5 Multiplexer and demultiplexer 6.6 Encoder and Decoder, 6.7 ROM and PLA
- Distinguish between synchronous and asynchronous logic and latch and flip flops. - Give knowledge of different flip flops and their use in sequential logic circuits. - Help to design and analyze sequential logic circuits.	Unit 7: Sequential Circuits (6 hrs) 7.1 Synchronous and asynchronous logic 7.2 Differences between Latch and fli-flop, Flip flops (RS, JK, D, T) and their truth table, excitation table and characteristic equation 7.3 Triggering of flip flops 7.4 State diagram and state table 7.5 State reduction and binary assignment 7.6 Design and analysis of clocked sequential circuit 7.7 Master-slave flip flops
- Describe about registers, shift registers and types with timing sequences. - Design synchronous, asynchronous and Mod	Unit 8: Registers and Counters (6 hrs) 8.1 Register, shift register and types of Shift register 8.2 Synchronous counters 8.2.1 up to 4-bit counters

counters.	8. 3 Asynchronous counters 8.3.1 BCD ripple counter, 8.3.2 Mod counter 8.4 Ring counter 8.5 Output hazard race
- Describe about read and write operation in RAM. - Design arithmetic circuit and logic unit. - Describe about the processor unit and its diagram.	Unit 9: Memory Unit and ALU (6hrs) 9.1 Random access memory 9.2 Design of arithmetic logic unit 9.3 Accumulator 9.4 Shifter and status register 9.5 Processor unit

5. Laboratory work

- Familiarization with logic gates.
- Familiarization with Boolean functions.
- Design of simple combinational circuits.
- Adder and subtractor
- Encoder and decoder
- Multiplexer and demultiplexer
- Design of flip flops.
- Registers and counters

6. List of Tutorials

The various tutorial activities that suits this course should cover all the content of this course to give student a space to engage more actively with the course content in the presence of instructor. Students should submit tutorials as assignments or class-works to the instructor for evaluation. The following tutorial activities of 15 hours should be conducted to cover all the content of course:

A. Discussion-based Tutorials: (3 hrs)

- Analog and digital signal and system.
- IC technology and parameters considered during fabrication
- Sequential circuits and types
- Output hazard races

B. Problem solving-based Tutorials: (12 hrs)

- Conversion of numbers among number systems
- Simplification of Boolean functions in SOP and POS using theorems and postulates.
- Simplification of Boolean function in SOP and POS using K Map.
- Logic gate implementation, used of NAND and NOR gates.
- Implementation of Boolean function using various MSI and LSI components.
- Design of various combinational circuits, code conversion circuits and parity generation and checking circuits.
- Design of synchronous sequential circuit from state diagram.
- Design of synchronous and asynchronous counters.
- Design of arithmetic and logic unit.

7. Evaluation system and Students' Responsibilities

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, term-exams, lab reports and projects etc. The tabular presentation of the internal evaluation is as follows:

External Evaluation	Marks	Internal Evaluation	Weight	Marks
Semester-End examination	50	Theory		30
		Attendance & Class Participation	10%	
		Assignments	20%	
		Presentations/Quizzes	10%	
		Term exam	60%	
		Practical		20
		Attendance & Class Participation	10%	
		Project Report	10%	
		Viva	20%	
		Exam	60%	
		Total Internal		50
Full Marks: 50 + 50 = 100				

Student Responsibilities

Each student must secure at least 45% marks in internal evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Books and References

Text Book

1. M. Morris Mano, *Digital Logic and Computer Design* Pearson India, 2017.

Reference Books

1. M. Rafiquzzaman, Steven A. McNinch, *Digital Logic*, John Wiley and Sons, 2019.
2. M. Morris Mano, *Digital Design*, Prentice Hall of India, 1998.

Pokhara University
Faculty of Science and Technology

Course No.: xxx xxx

Course title: Electronics Device and Circuits (3-1-2)

Nature of the course: Theory & Practical

Level: Undergraduate

Full marks: 100

Pass marks: 45

Time per period: 1 hour

Total Periods: 45

Program: BE

1. Course Description

This course covers an introduction to active electronic components and a study of circuits containing such components. This is designed to teach the students about theory, concepts and principles of operation of various electronics devices related to their use and working in electronic systems and applications. It also aims to introduce students the analysis and design techniques of circuits involving these discrete devices as well as the integrated circuits.

2. General Objectives

The general objective of the course are:

- To equip the students with in-depth concepts and the principles of operation, construction and characteristics of semiconductor devices, and their utilization in basic electronics building blocks (or modules) and their performances.
- To provide the students with the techniques of analysis and design of basic building blocks of modern technology using devices.
- To acquaint the students with diode as a rectifier, feedback and design feedback amplifier, and oscillators & power amplifiers using transistor.

3. Methods of Instruction

Lecture, discussion, Tutorials, Assignments, Lab works, Project works, Presentation

4. Course in Detail

Specific Objectives	Contents
• explain the basic function of diodes in electrical circuits and describe the characteristics of an ideal diode; • describe the electrical characteristics of conductors, insulators and semiconductors; • discuss the doping of semiconductor materials and the construction of semiconductor diodes; • describe the characteristics of a typical diode and sketch its current-voltage characteristics;	Unit I: Semiconductor diode (6 hrs) 1.1 Review of insulator and semiconductor and conductors 1.2 Conduction in semiconductors 1.3 Theory of p-n junction 1.4 Forward and reverse biasing of diode 1.5 Diode as a nonlinear device 1.6 Ideal and piecewise linear model of diode 1.7 The effects of temperature in V-I characteristic curves 1.8 Junction capacitances and its effects 1.9 Diode switching times 1.10 Junction breakdown

- outline the use of several forms of special-purpose semiconductor devices, including Zener, tunnel and varactor diodes; - Design a range of circuits that exploit the characteristics of semiconductor diodes.	1.11 Construction, characteristics and applications of Zener diode, Schottky diode
- explain the importance of bipolar transistors in modern electronic circuits; - describe the construction, operation and characteristics of bipolar transistors; - analyze simple amplifier circuits based on transistors and determine their operating conditions and voltage gain; - discuss the importance of negative feedback in overcoming variability in such circuits; - describe the low-frequency behavior of amplifiers that use coupling capacitors; - List a range of applications of bipolar transistors in addition to their uses in amplifiers.	Unit II: Bi-polar Junction Transistor (6 hrs) 2.1 Introduction of bipolar junction transistor 2.2 Current flow mechanism in PNP and NPN transistors 2.3 Input and output characteristics of CE and CB transistor amplifiers 2.4 Reach through and punch through effects 2.5 Active and cut off and saturation modes of operations of BJT and BJT switching times 2.6 The transistor as an amplifier and switch 2.7 Comparison of CB, CE and CC configurations 2.8 BJT biasing, dc load line, ac load line and Q point 2.9 Stability factor
- Explain and analyze the operation of both half and full wave rectifiers - Explain and analyze filters and regulators and their characteristics - Explain and analyze the operation of diode limiting and clamping circuits	Unit III: DC Power supply (5 hrs) 3.1 Half wave and full wave rectifiers 3.2 Average value, RMS value ripple factor of half and full wave rectifiers 3.3 Filtering process, Shunt capacitor and LC filter and pi filter 3.4 Series Shunt and biased clipper circuits 3.5 Clamping circuits 3.6 Regulated and unregulated power supplies 3.7 Transistor series and transistor shunt regulators
- list the various major forms of field-effect transistor (FET); - describe the characteristics common to all forms of FET and explain how these characteristics make them suitable for use in amplifiers; - describe the physical operation of both MOSFETs and JFETs and explain how this influences the characteristics of these devices; - outline the behavior of FETs;	Unit IV: The Field Effect Transistor (FET) (5 hrs) 4.1 Comparison between FET and BJT 4.2 Construction and working principle of JFET 4.3 Biasing and load line 4.4 Drain and transfer characteristics of JFET and JFET parameters 4.5 Construction and working principles of DMOSFET and EMOSFET

• Develop the small signal models of transistor that are used in analysis of linear amplifier. • Small signal hybrid model- measurement and analysis using h-parameter • Low frequency r_e model and analysis using r_e model	Unit V: The Small Signal Low Frequency Analysis Model of BJT (5 hrs) 5.1 Low frequency hybrid model 5.2 Transistor configurations and their hybrid model- Measurement of h-parameters and analysis of a transistor amplifier circuit using h-parameters 5.3 Low frequency r_e model, amplifier configuration and their expression for voltage gain, current gain, input impedance and output impedance using r_e model 5.4 Analysis of transistor amplifier circuit using r_e model 5.5 Emitter follower
• Describe methods for inter-stage coupling in multi-stage amplifiers: • Direct Coupling • Capacitor coupling. • Transformer Coupling	Unit VI: Multistage Amplifiers (4 hrs) 6.1 Multistage amplifier and Gain calculation of n-stages cascaded amplifiers 6.2 Methods of coupling 6.3 Expression of voltage gains, Current gains, input and output impedance for two stages RC coupled amplifier using r_e model 6.4 Choice of configuration in a cascade 6.5 Darlington-pair amplifier and its effective beta
• Different classes of operation of Amplifiers • Power conversion capabilities and applications • Merits and demerits • Problems with distortion in Amplifiers and remedies • Push-Pull power Amplifiers • Advanced Power Amplifiers	Unit VII: Large Signal Amplifiers (4 hrs) 7.1 Analysis of large signal model 7.2 Class A, B, AB and Class C amplifiers 7.3 Push-pull amplifiers 7.4 Cross over distortion 7.5 Transformer coupled push-pull stages 7.6 Amplifier efficiency, power dissipation and heat sinks
• Principle of Feedback Amplifiers • Advantages of Negative feedback • Gain stability • Increased bandwidth	Unit VIII: Feedback Amplifiers (3 hrs) 8.1 Negative feedback amplifiers and advantages of negative feedback 8.2 Gain stability, extension of bandwidth 8.3 Importance of positive feedback on oscillation
• Familiarity with operational amplifiers (op-amps) and basic op-amp circuits: • Inverting, non-inverting, and summing amplifiers.	Unit IX: Operational Amplifier and Oscillator (7 hrs) 9.1 Basic Model 9.2 Ideal and non-ideal properties 9.3 Virtual ground concept, offset voltage, input bias current, slew rate and CMRR

Input and output impedances of op-amp circuits.	9.4 Inverting and non-inverting amplifier 9.5 Integrator, differentiator and summing amplifier and their applications 9.6 Astable and monostable multivibrators 9.7 Barkhausen criteria for oscillation 9.8 RC phase shift and Wein bridge oscillator using Op-amp
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5. Laboratory Work

- Study of V-I characteristics of PN diode and Zener diode.
- Study of half wave and full wave rectifiers.
- Study of input and output characteristics of CE and CB transistor amplifier.
- Measurement of gain in single stage and multistage amplifiers.
- Frequency response of CE amplifier
- Frequency response of CC amplifier
- Measurement of efficiency of class A and Class B push pull power amplifiers.
- Design of RC phase shift and Wein Bridge oscillator.
- Measurement of Regulation in series regulator against change in input voltage and load resistance.
- Study of drain characteristics of JFET.

6. List of Tutorials

The various tutorial activities that suits your course should cover all the content of the course to give students a space to engage more actively with the course content in the presence of Instructor/professor. The following tutorial activities of 15 hrs. should be conducted to cover the content of this course:

- Discussion-based Tutorials: (2 hrs)
- Numerical and analysis based numerical of diode, Zener diode, and transistor. (5 Hrs)
- Discussion and designing of biasing circuit (2 Hrs)
- Analysis and numerical solution of small and large frequency signal amplifier. (3Hrs)
- Numerical problems for operational and feedback amplifier. (3Hrs)

7. Evaluation System and Students' Responsibilities

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, term-exams, lab reports and projects etc. The tabular presentation of the internal evaluation is as follows:

External Evaluation	Marks	Internal Evaluation	Weight	Marks
Semester-End examination	50	Theory		30
		Attendance & Class Participation	10%	
		Assignments	20%	
		Presentations/Quizzes	10%	
		Term exam	60%	
		Practical		20

		Attendance & Class Participation	10%	
		Project Report	10%	
		Viva	20%	
		Exam	60%	
		Total Internal		50
Full Marks: 50 + 50 = 100				

Student Responsibilities

Each student must secure at least 45% marks in internal evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Books and References

Text Books

1. Jacob Millman & Christors C. Halkias, Electronic Devices and Circuits, Tata McGraw Hill, India.
2. Theodore F. Bogart, Electronic Devices and Circuits, Univesal Book Stall, Indial.

References

1. Roberst Boylestad & Louis Nashelsky, Electronic Devices and Circuit Theory, Prentice Hall, India.
2. Allen Mottershead, Electronic Devices and Circuits, Prentice- Hall, India
3. Albert Paul Malvino, Electronic Principles, Tata Mc Graw Hill, India.
4. S. Sedra and K.C. Smith, Microelectronic Circuits, Holt, Rinehart and Inc., New York.

Pokhara University
Faculty of Science and Technology

Course No.: xxx xxx
Course title: Programming in C (3-1-3)
Nature of the course: Theory & Practical
Level: Undergraduate

Full marks: 100
Pass marks: 45
Time per period: 1 hour
Total Periods: 45
Program: BE

1. Course Description

This course first introduces the different computer/programming languages- low-level and high-level languages and generation of the programming languages. This course enables the students to solve a given problem using a structured programming language- the C language through the problem-solving steps- problem analysis, design of algorithm, flowcharting, coding the algorithm, executing and compiling/interpreting the program, testing and debugging the program and finally well documenting the program for future understanding. The aim of this course is to provide the basic knowledge of structured programming concepts to develop a program using the C language.

2. General Objectives

The general objectives of this course are:

- To provide the basic knowledge of computer/programming languages and generations of programming languages.
- To provide the basic knowledge of problem-solving steps to resolve a particular problem using a computer program.
- To provide the basic concepts of structured programming with the C language.

3. Methods of Instruction

3.1.General instructional Techniques: Lecture, discussion, readings.

3.2.Specific instructional Techniques: Lab works, Project works

4. Contents in Detail

Specific Objectives	Contents
• familiar with the various types of programming languages, programming paradigms and software • analyze a problem, find the solution, implement the solution using a particular computer/programming language and finally prepare its documentation	Unit I: Programming Languages and Problem Solving (6 hrs) 1.1 Programming Languages- machine-level language, assembly language and high-level language. 1.2 Software and its types 1.3 Generations of programming languages 1.4 Structured programming 1.5 Problem solving using computer- Problems Analysis (understanding of the problem, feasibility and requirement analysis), Design (Algorithm and flowchart), Coding (compilation/interpretation and execution),

	Testing and debugging, Implementation, Evaluation and Maintenance of computer programs, Program documentation
- explain the history and features of C programming, implement the various constructs of C programming like constants, variables, keywords, data types, operators and formatted I/O functions etc. - write and evaluate the expressions in C programs	Unit II: Introduction to C (6 hrs) 2.1 History of C language 2.2 Features of C 2.3 The C as a middle-level language 2.4 The C as a system programming language 2.5 The C character set 2.6 Keywords and Identifiers 2.7 Data types 2.8 Constants, variables and their declaration 2.9 Formatted input/output functions 2.10 The C Operators
implement the various control structures of C programming with reference to algorithm or flowchart they developed already	Unit III: Control Structures (6 hrs) 3.1 Introduction and types of control statements- sequential, branching and looping statements 3.2 Branching statements- simple if statement, if-else, nested if, if-else-if ladder and switch statements 3.3 Looping statements- for loop, while loop, do-while loop, nested loop 3.4 The break, continue and goto statements
- represent/store homogenous type of data as array and realize the multi-dimensional array representation - implement string handling functions to handle textual data	Unit IV: Arrays and Strings (6 hrs) 4.1 Introduction to arrays 4.2 One dimensional and Multidimensional arrays 4.3 Initialization of arrays and accessing the elements of arrays 4.4 Strings- the character arrays 4.5 Functions related to the strings
- use function to perform certain operation - use macro definition, header files and function prototype and implement them in their program	Unit V: Functions (6 hrs) 5.1 Introduction 5.2 Importance of functions 5.3 Returning a value from a function and sending a value to a function 5.4 Function prototypes 5.5 Calling a function- Call by value 5.6 Recursive functions 5.7 Passing an array to a function 5.8 Local variables, formal parameters and global variables 5.9 Storage classes 5.10 Pre-processor directives- C libraries, macros and header files
implement the pointers in program	Unit VI: Pointers (6 hrs)

allocate and deallocate memory dynamically in C language	6.1 Introduction 6.2 Pointer operators 6.3 Pointer arithmetic 6.4 Returning multiple values from functions using pointers 6.5 Pointers and Arrays 6.6 Double indirection 6.7 Dynamic memory allocation
able to understand how the heterogeneous data are represented and stored in structure and union and implement them in program	Unit VII: Structure and Union (5 hrs) 7.1 Definition of Structure 7.2 Nested-Structure 7.3 Array of Structure 7.4 Structures and Pointers 7.5 Union 7.6 Self-referential structure
create files, access the contents of the file and perform the required operations in the file.	Unit VIII: Files and File Handling (4 hrs) 8.1 FILE pointer, File opening modes (read, write, append) 8.2 File handling functions 8.3 Creating and operating a file in different modes

5. List of Experiments

Laboratory works should cover all the concepts of C language studied in the lectures. Students should submit a final project that uses all the constructs and features of C studied in this course. The marks for the practical work will be based on the project work.

Tools: any C/C++ compiler

6. List of Tutorials

The various tutorial activities that suit this course should cover all the content of this course to give students a space to engage more actively with the course content in the presence of instructor. Students should submit tutorials as assignments or class works to the instructor for evaluation. The following tutorial activities of 15 hrs should be conducted to cover all the content of this course:

A. Discussion-based Tutorials: (2 hrs)

1. Evolution of Programming languages and its generations (Class discussion)
2. Software and its types.
3. Generations of programming languages.
4. Structured programming. (Oral Presentation).

B. Problem solving-based Tutorials: (10 hrs)

5. Develop algorithms and flowcharts to solve various problems such as to find largest number among three numbers, prime numbers, temperature conversion, product of matrices, finding sum of the terms in series, printing various patterns etc.
6. Develop the C programs for the problems for which you developed the algorithms.
7. Write a program to pass an array to a function.
8. Write a program to use pointers to pass multiple values from a function.

9. Write a program to use the basic string functions to manipulate string data.
10. Write a program to use the principle of recursion to solve the complex problems such as to find factorial of a number, fibonacci series.
11. Write a program to illustrate the macros and header files.
12. Write a program to illustrate how memory is allocated and deallocated in C language.
13. Write a program to use the nested structure. Discuss the scenarios when the structures and unions are used in real practice.
14. Write a program to solve simple file handling problems.
- C. Review and Question/Answer-based Tutorials: (3 hrs)
15. Case study of “Development of C with the UNIX operating system and origin of C++ languages” followed by Oral Presentation in class.
16. Students ask questions within from the course content and assignments and review key course content in preparation for tests or exams.

7. Evaluation System and Students’ Responsibilities

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, test-exams, term-exams, lab reports and projects etc. The tabular presentation of the internal evaluation is as follows:

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		Presentations/Quizzes	10%	
		Term exam	60%	
		Practical		20
		Attendance & Class Participation	10%	
		Project Report	10%	
		Viva	20%	
		Exam	60%	
		Total Internal		50
Full Marks: 50 + 50 = 100				

Student Responsibilities

Each student must secure at least 45% marks in internal evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Books and References

Text Books

1. Gottfried, B. *Programming with C*. New Delhi: McGraw-Hill.

2. Balagurusamy, E. *C programming*. New Delhi: McGraw-Hill.

References

1. Kelley A. & Pohl, I. *A Book on C, Programming in C*, USA: Addison-Wesley.
2. Kernighan B. W. & Ritchie, D. M. *The C Programming Language*. USA: Prentice Hall Publication.